MIL-H-52079B(AT)
2 May 1969
SUPERSEDING
MIL-H-52079A(ORD
23 September 1959

MILITARY SPECIFICATION

HOSE, PREFORMED: SEMI-FLEXIBLE, REINFORCED

1. SCOPE

1.1 Scope. This specification covers requirements for one type of semi-flexible preformed hose, for air induction on heavy duty engines.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, for a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-W-461 - Wire, Steel, Carbon (Round, Bare, and Coated).

Military

MIL-H-775 - Hose, Rubber or Fabric (Including Tubing); and Fittings, Nozzles, and Strainer Packaging of.
MIL-F-13927 - Fungus Resistance Test; Automotive Components, General Specification for.

STANDARDS

Federal

FED. TEST METHOD STD. NO. 601 - Rubber; Sampling and Testing.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-417 - Rubber Compositions, Vulcanized, General Purpose Solid (Symbols and Tests).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Hose furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.4).

3.2 Materials. Unless otherwise specified herein, materials shall conform to applicable drawings, standards, and specifications.

3.2.1 Brand and identification marking. The manufacturer's brand or firm name, date of manufacture, quarter and year, and part number shall be marked on the exterior of the hose. The marking shall be in a contrasting color, shall be legible, and shall serve to identify the product.

3.2.2 Date of manufacture. The hose shall be not more than 12 months old when it is submitted for inspection.

3.3 Design and construction. Design and construction of hose furnished under this specification shall conform to applicable drawings (see 6.2).

3.4 Performance.

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3.4.1 Bending. When tested as specified in 4.5.2, the hose shall evidence no kinking, flattening, fracturing, or deformation.

3.4.2 Collapsing resistance. When tested as specified in 4.5.3, after release of the load, which shall be applied for 5 minutes, no portion of the hose shall show evidence of a permanent set, distortion, or deformation in excess of 5 percent in diameter.

3.4.3 Accelerated-aging flexibility. When tested as specified in 4.5.4, hose shall show no evidence of roughing of compound, rupture, cracks, bulging, separation, checks, or leakage.

3.4.4 Pulsating proof pressure. When tested as specified in 4.5.5, hose shall not bulge, separate, or show evidence of leakage.

3.4.5 Burst strength. When tested as specified in 4.5.6, hose shall show no evidence of bursting or leakage, at any pressure up to and including 75 pounds per square inch (p.s.i.).

3.4.6 Fungus resistance. After being tested as specified in 4.5.7, hose shall meet the requirements specified in 3.4.5.

3.4.7 Friction (adhesion). When tested as specified in 4.5.8, the force required to separate adjacent parts shall be not less than 12 pounds, tube to ply.

3.4.8 Resistance to vacuum. When tested as specified in 4.5.9, the outside diameter shall not be reduced during the 5 minute test period specified, while subjected to a vacuum of 20 inches of mercury, the inner tube shall not collapse or separate from the reinforcement. Separation of the tube from reinforcement may be determined by sectioning the hose lengthwise.

3.4.9 Metal flanges. Where metal flanges are used, hose shall be bonded to a metal "V" type flange on either or both ends. The bond of the hose to the metal flange shall withstand a tensile load of 300 pounds without separating. (see 4.5.10).

3.5 Workmanship. Workmanship shall be such as to produce a hose which will be free from imperfections such as cuts, breaks, blisters, exposed reinforcements, pitting, and rough surfaces within the AQL's of section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification inspection. A qualification sample consisting of 15 hoses of each design configuration (flanged or plain), each 36 inches long and having a nominal I.D. of 4-1/2 inches shall be submitted for qualification inspection (see 6.2). The samples shall be representative of the hose proposed to be furnished under the contract. Qualification inspection shall be performed by the Government and shall consist of examination for the defects specified in 4.3.2.2 and testing as shown in table I, in the order given.

Table I. Qualification tests.

Sample	Test No.	Type of test
1	4.5.3	Collapsing
3	4.5.4	Accelerated aging
	4.5.8	Friction
	4.5.9	Vacuum
1	4.5.2	Bending
2	4.5.5	Pulsating
2	4.5.6	Burst
3	4.5.7 and 4.5.6	Fungus resistance
1	4.5.10	Flange tensile test

4.2.1 Failure. Failure of a qualification sample to pass any of the examinations or tests specified herein may be cause for the Government to refuse to conduct additional testing until the faults revealed by the tests have been corrected.

4.2 Quality conformance inspection.

4.3.1 Sampling.

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4.3.1.1 Lot formation. A lot shall consist of all hose of one type and size from an identifiable production period, submitted at one time for acceptance.

4.3.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with inspection level II of MIL-STD-105.

4.3.1.3 Sampling for testing. Samples for quality conformance testing shall be selected in accordance wit inspection level S-4 of MIL-STD-105.

4.3.2 Quality conformance examination.

4.3.2.1 Acceptable quality level. Each sample selected in accordance with 4.3.1.2 shall be examined for conformance to the following acceptable quality levels (AQL), on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.3.2.2 Classification of defects. For examination purposes, defects shall be classified as specified in table II.

Table II. Classification of defects.

Categories	Defects	Method of inspection
Major:		
101	Construction not as specified (see 3.3)	Visual
102	Dimensional nonconformance (see 3.3)	Gage
103	Poor workmanship (see 3.5)	
Minor:		
201	Improper identification (see 3.2.1)	Visual
202	Improper finish (see 3.5)	Visual

4.3.3 Testing.

4.3.3.1 Quality conformance tests. Samples selected in accordance with 4.3.1.3 shall be subjected to the tests specified in 4.5.2 and 4.5.3, using an AQL of 6.5 on the basis of percent defective.

4.4 Control tests.

4.4.1 Sampling for control tests. Hose or hose assemblies shall be selected at the rate of 5 per month, o 5 from each 5,000 produced, except that not more than 10 shall be selected in any given 30-day period. The sam ple shall be identified as to production period, examined for the defects specified in 4.3.2.2, and subsequently su jected to tests specified in 4.5.6, 4.5.8, and 4.5.9.

4.4.2 Failure. Failure of a control test sample to pass any specified examination or test may be cause fo the Government to refuse to accept subsequent lots until it has been proved to the satisfaction of the Government that the faults revealed by the tests have been corrected.

4.5 Test procedures.

4.5.1 Conditions and equipment. Except where otherwise specified herein, tests shall be made in an ambient air temperature of 77° ± 10°F on unaged specimens. Specimens tested at elevated or reduced temperatures shall have been temperature-stabilized or kept for the specified time, in temperature-controlled cabinets at spec fied temperatures.

4.5.2 Bending test. To determine conformance to 3.4.1, hose shall be installed in a contour-forming fixture and subjected to a minimum radius bend of 1-1/2 times the inside hose diameter for each of 4 separate direc tions spaced 90 degrees apart.

4.5.3 Collapsing resistance test. To determine conformance to 3.4.2, any 6-inch longitudinal section sha be subjected to a load of 125 pounds uniformly applied. After release of load, the diameter shall be measured an any changed recorded.

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4.5.4 Accelerated aging flexibility test. To determine conformance to 3.4.3, the hose shall be subjected to ie test specified in method 7221 of FED. TEST METHOD STD. NO. 601,except that the temperature shall be 250° 3.6°F for a period of 24 + 1/2 hours and upon completion shall be subjected to a temperature of -65° + 2°F for a eriod of 24 hours, and while still at the latter temperature to be flexed immediately through 5 cycles of 10 secnds each, over a mandrel having a diameter 10 times the inside hose diameter. Each flexing cycle shall consist f generating hose contact with the mandrel diameter form progressively from one hose end to the other, then reolving the hose bottom side up, and repeating the hose contact generation starting the initial contact from the opposite end. After flexing, the hose shall be subjected to the test specified in 4.5.5, and re-examined for conformnce to 3.4.3.

4.5.5 Pulsating proof pressure. To determine conformance to 3.4.4, the hose shall be tested as specified in iethod 10211 of FED. TEST METHOD STD. NO. 601, except for quality conformance testing, lengths shall be as pecified on applicable drawings, and the pressure shall be 40 p.s.i. applied internally at pulsating intervals of 5 econds "on", 3 seconds "off" for 15 minutes.

4.5.6 Burst strength. To determine conformance to 3.4.5, the hose shall be tested as specified in method 0011 of FED. TEST METHOD STD. NO. 601.

4.5.7 Fungus resistance. To determine conformance to 3.4.6, the hose shall be subjected to class 3, methd A of MIL-F-13927, sample No. 1 for 30 days, sample No. 2 for 60 days, and sample No. 3 for 90 days, and pon completion shall be subjected to the test specified in 4.5.6.

4.5.8 Friction (adhesion). To determine conformance to 3.4.7, hose specimens shall be subjected to the riction test specified in method 8011 of FED. TEST METHOD STD. NO. 601, except that exposed wire ends shall ot be considered when preparing test sample.

4.5.9 Vacuum test. A 36-inch length of hose shall be assembled with end couplings. The outside diameter f the hose shall be measured and recorded. One end shall be closed in such a manner as to prevent leakage of ir, and the other end shall be connected to a vacuum pump. The hose shall be subjected to a vacuum of 20 inches f mercury while curved to a radius equal to 5 times the diameter of the hose. After a period of 5 minutes, and ith vacuum maintained, the hose diameter shall again be measured and the hose inspected for conformance to .4.8.

4.5.10 Flange tensile test. Hose having metal flanged ends shall be tested to determine conformance to .4.9.

4.6 Inspection of preparation for delivery. Inspection of preservation, packaging, packing, and marking hall be in accordance with the applicable procedures outlined in MIL-H-775 (see 5.1).

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing, and marking shall e in accordance with the requirements of MIL-H-775 for the applicable levels of protection specified (see 6.2).

6. NOTES

6.1 Intended use. The hose covered by this specification is intended for use in heavy-duty military vehicle ngines such as air cleaner to carburetor application, where a material having good resistance to normal engine eat and to motor oil dripping or splashing is required. It is not intended for resistance to constant, high heat operating level approximately 200°F) nor for a purpose requiring high resistance to oil or having permanent conact with oils.

NOTE:

Hose made from materials having the following characteristics has been found satisfactory:

NYLON:

Weight	2.30 ounces per sq. yd.
Size	0.0065 gauge
Minimum tensile strength	ASTM Grab
	Warp: 146
	Fill: 155

RUBER:

Synthetic rubber compound grade SC-710 A_1, B_1, C_1, F_2 of MIL-STD-417.

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WIRE:

Wire, Grade FS-1035 or FS-1045, Composition A, finish 2, QQ-W-461.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of applicable drawings (see 3.3).
(c) Where qualification samples shall be sent (see 4.2).
(d) Selection of applicable levels of preservation, packaging, and packing (see 5.1).

6.3 Special provisions for establishing qualification. Qualification of hose of 4-1/2 inches I.D. submitted by a manufacturer will be recognized as establishing qualification of all sizes covered by this specification, submitted by the same manufacturer.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for the opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the U.S Army Tank-Automotive Command, Warren, Michigan 48090 and information pertaining to qualification of products may be obtained from that activity.

Custodian:
Army - AT

User activity:
Army - ME

Preparing Activity:
Army - AT

Project No. 2940-A027

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-H-52079B(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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U.S. Army Tank-Automotive Command
Warren, Michigan 48090
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